COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithms	**SECTION:**	CPE 21S2
DATE PERFORMED:	August 26, 2026	**DATE SUBMITTED:**	August 26, 2026
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Prelim Exam **SET D**

1. PROCEDURE OUTPUT

Algorithm:

1.) Create a constructor and destructor for array based stack or linked list based queues
2.) Prompt positive integers to determine whether the sequence is symmetric or not
3.) Determine whether the given inputs are symmetric or not on each integers
4.) Enqueue the integers into the queue
5.) Pop the old integer in order to stack a new integer
6.) Dequeue the front integer of the queue
7.) Compare both dequeue and the popped value if its symmetrical or not
8.) If the integers are not the same then turn the symmetric to false
9.) If the integers are equal then set the symmetric to true
10.) Display the output of the integers along with the judgement of the output if it is “Symmetrical” or “”Not Symmetrical”

Pseudocode:
START the program
INPUT the sequences of integers
DETERMINE whether the given inputs of sequences and integers are symmetrical or not symmetrical
IF its symmetrical make it true
IF its not symmetrical make it false
DISPLAY the input sequence and if its symmetrical or not
END of program

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic

work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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